



FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

UNIVERSITI TUNKU ABDUL RAHMAN (UTAR)

Course Code: UCCD3084

Course Title: Graphics Programming for Extended Reality (XR)

Trimester: June 2026

DESIGN REPORT

Project Title: Factory VR Training Simulation: Repair, Optimization & Custom Feature

Team Name: Group 1

Submission Date: 29/8/2026, Saturday

Group Members:

Student ID	Student Full Name
2103940	Yeo Pei Han
2304963	Tee Ing Harn
2206770	Leong Sum Yue
2206054	Teoh Wei Zhong

1.0 Introduction

Virtual Reality (VR) offers an immersive and interactive experience that allows users to engage in training activities within a virtual environment. This project is about fixing and upgrading an existing VR factory training prototype created in the Unity game engine and XR Interaction Toolkit. There are also a number of UI/UX, interaction, lighting and visual feedback issues with the original prototype that need to be fixed. In addition to addressing these problems, the project enhances the overall VR experience by adding locomotion, comfort settings, spatial audio and resets. An interactive training feature is also developed to expand and enhance the current factory environment, giving users a comprehensive VR training experience.

1.1 Project Scope

The goal of this project is to repair, improve and extend the supplied VR factory training prototype in Unity and XR Interaction Toolkit. The factory environment is already affected by seven issues which are the focus of the first part of the project. This includes correction of onboarding fade screen to be able to cover the user's view, the Console Labels toggle is now hidden only the labels and the baked lighting of the factory floor, box piles, and control console are enhanced. The magnet interaction is also enhanced with more visual feedback, with the magnet going green when it is switched on, and further visual feedback when lifting a box. The interaction during onboarding is also updated so that the user is not able to interact with the factory controls until the instructions are followed. Users can move around the factory with teleportation, and the snap turning and vignette settings offer additional comfort features. Spatial audio is integrated, as it is used to create sounds with more responsive behavior in the virtual space based on the user's location. Additionally, there is a Reset function implemented to reset the training environment back to default without the need to load the entire scene. Last but not least, Part B introduces a new training component to the current prototype which is interactive VR training. This feature consists of meaningful interaction, correct and incorrect action feedback, contextual UI guidance, visual effects and comfort friendly interaction design. The project aims to enhance the functionality, usability, visual quality and immersion of the VR factory, while maintaining its primary structure.

2.0 Part A

Part A focuses on identifying and repairing the existing issues in the provided VR factory prototype. The repairs cover user interface behaviour, interaction logic, lighting, and visual feedback while maintaining the original structure and intended functionality of the factory environment. Figure 2.0.1 provides an overview of the seven repair areas completed in Part A.

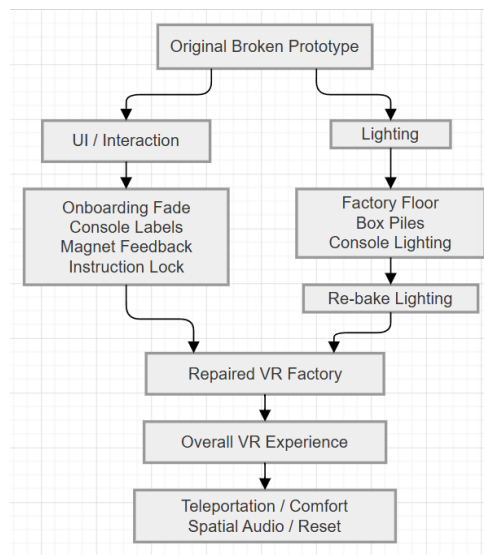


Figure 2.0.1: Part A Repair Overview

2.1 UI — Onboarding Fade Screen

The first repair was the onboarding fade screen that has been put in the middle of the factory in the original prototype rather than covering the user's visual field. This made this transition less effective because the black fade was meant to block the user's view for a short time during the onboarding process. The solution was found by examining the fade screen and its Canvas setup, and tweaking them to ensure that the UI element is displayed at the correct position in relation to the user's VR view. The placement and display behaviour of the fade element were corrected so that it appears in front of the user and covers the visible area instead of appearing as an ordinary object within the factory environment. The black fade screen that is triggered correctly covers user's field of view after the modification and allows for a smoother fade into the VR training environment. This prevents distracting parts of the factory from remaining visible during the transition and provides a smoother onboarding experience for the user.

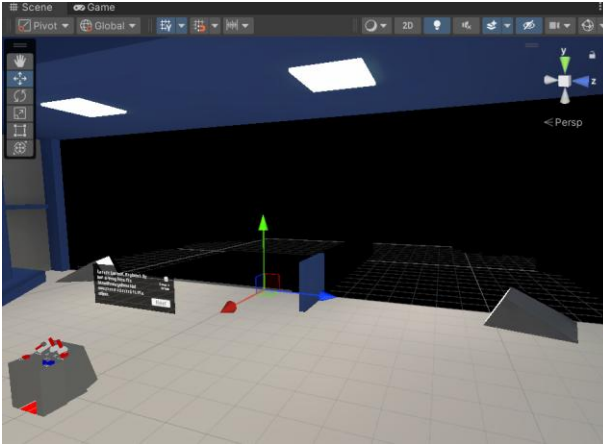


Figure 2.1.1:

Onboarding fade screen before repair

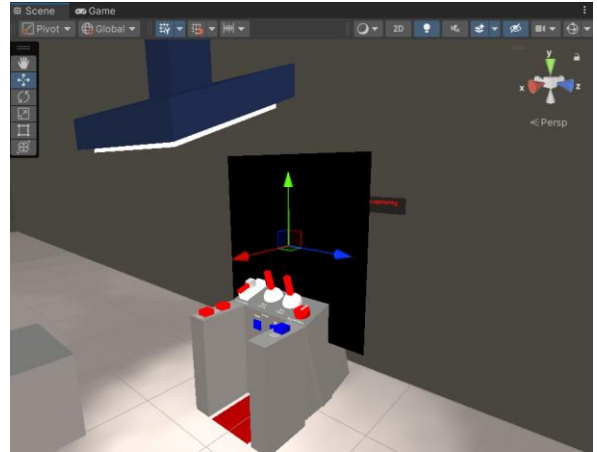


Figure 2.1.2:

Onboarding fade screen after repair

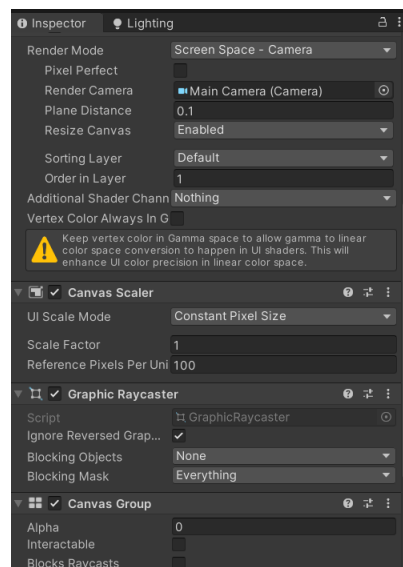


Figure 2.1.3: Fade screen Canvas (Plane Distance) configuration in Unity

2.2 Toggle Logic — Console Labels

The second repair fixed the misbehaviour of the Console Labels toggle. The "Console Labels" option, when disabled in the prototype, also removed the control console, but was meant to remove only the labels so that the console and the interactive controls remained. To resolve this issue, the object reference controlled by the toggle functionality was examined and modified to make the visibility change happen at the console label elements, instead of the parent console object. The toggle logic detaches the labels from the console, enabling the ability to enable or disable the labels

without changing the actual console or its interactive elements. After the repair, enabling the Console Labels option displays the labels for the different controls, while disabling the option hides only the labels. The console and its interactive controls remain visible and usable in both states. There is no way to hide/restrict the rest of the console in either state, it's always there and working. This modification gives the desired toggle functionality and enables the user to select if further textual guidance is to be provided, while at the same time not blocking the user from using the factory control system.

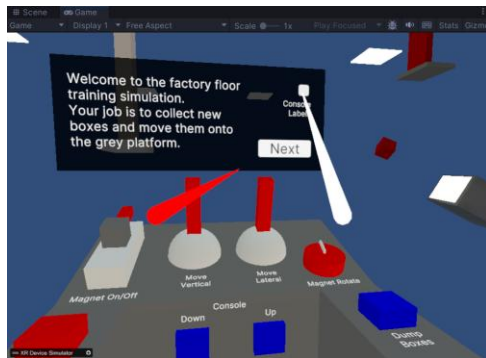


Figure 2.2.1:

Console Labels behaviour before repair

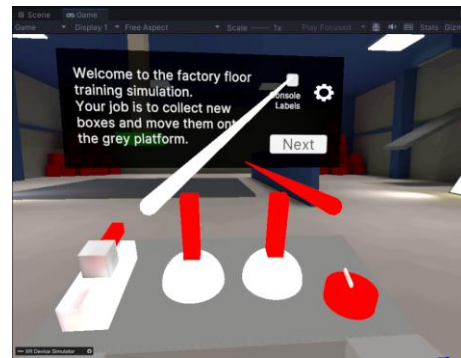


Figure 2.2.2:

Console Labels enabled after repair

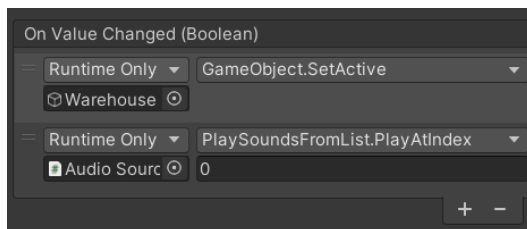


Figure 2.2.3:

Toggle configuration in Unity (Before)

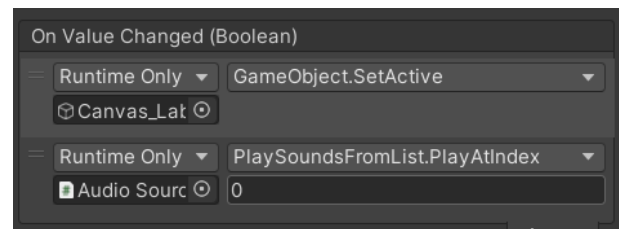


Figure 2.2.4:

Toggle configuration in Unity (After)

2.3 Lighting — Factory Floor

The third repair was about the baked-lighting quality of the factory floor. The original floor had an uneven appearance in the factory and lacked lighting details that contrasted with the other surfaces, creating an imbalance of texture within the factory. The lighting quality was checked by examining and adjusting the corresponding lightmap setup of the factory floor to provide an appropriate level of lightmap detail for the surface. The factory lighting was then re-baked with

the adjusted lightmap settings to recreate the lightmap information with the new settings. This will lead to baked lighting that will give more light and shadow detail on the factory floor and make it closer to the appearance of the surrounding surface. Another reason to use baked lighting in the VR environment is that all the lighting doesn't need to be calculated on the fly, it is calculated in advance. Hence, the repair takes into consideration the visual understanding of the factory floor and ensures a lighting approach that is optimized for the performance demands of a VR application.

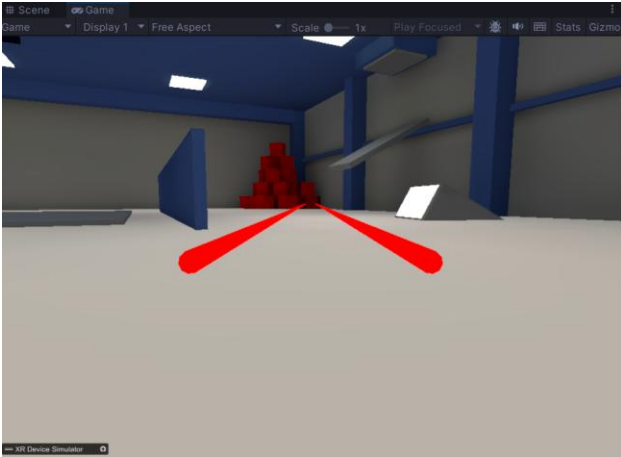


Figure 2.3.1:

Factory floor lighting before repair

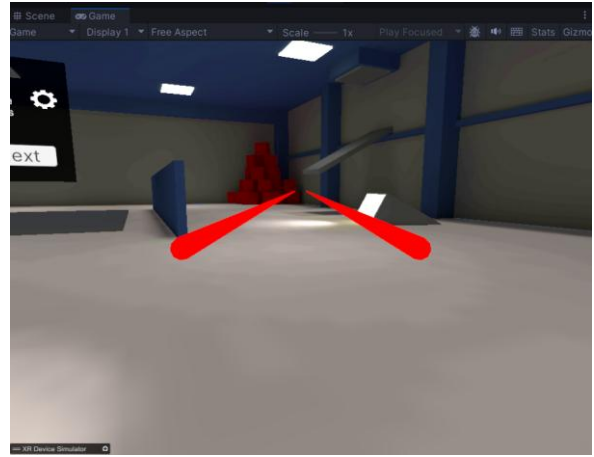


Figure 2.3.2:

Factory floor lighting after lighting re-bake

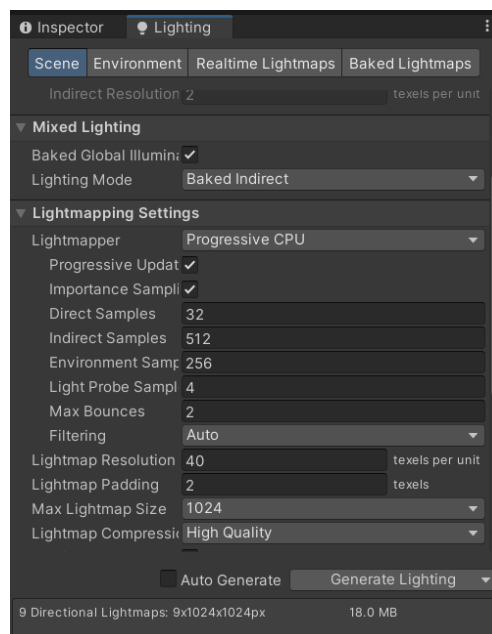


Figure 2.3.3: Factory floor lightmap configuration

2.4 Lighting — Box Piles

The fourth repair was the lighting and shadow quality of the box piles in the factory being baked. The pile of boxes in the back-left corner did not have realistic baked lighting and shadows as in other box piles, making it look out of place visually with the rest of the boxes. To solve this issue the affected box objects and lightmap baking settings were reviewed to verify that they were present in the baking-lighting process. The relevant lighting parameters were adjusted and the factory lights were then re-baked, and the lighting and shadow data for boxes were re-generated by Unity. The re-baking process resulted in the back-left box pile being lit and shaded more in keeping with the other box piles in the factory. This works to provide a more consistent look to the overall shot, since objects that are similar in one environment will behave the same way in the baked lighting. The repair also continues to keep the optimized baked-lighting technique necessary for the VR simulation, but adds to the realism of the factory environment.

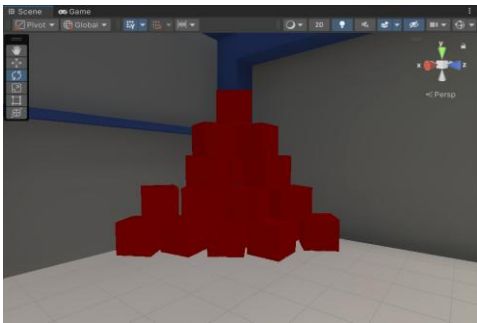


Figure 2.4.1:

Back-left box pile before lighting repair

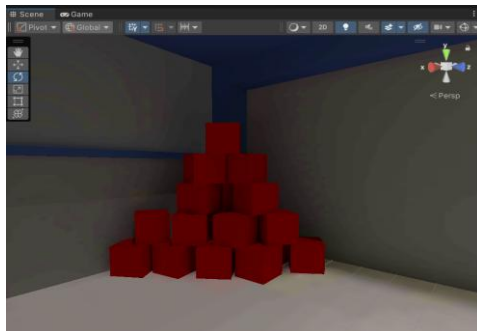


Figure 2.4.2:

Back-left box pile after lighting re-bake

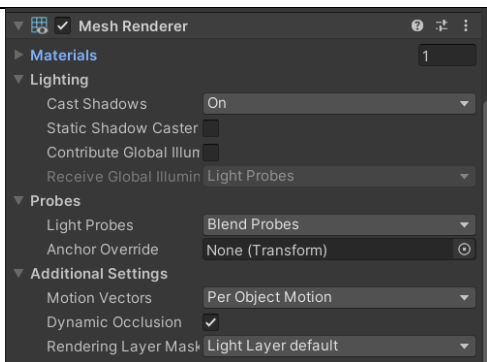


Figure 2.4.3:

Box Pile Lighting Configuration Before Repair

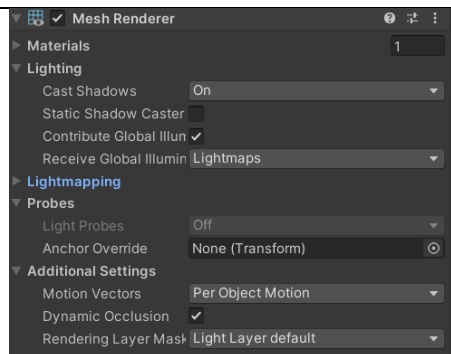


Figure 2.4.4:

Box Pile Lighting Configuration After Repair

2.5 Lighting — Console

The fifth repair was to lighten the factory control console. The original prototype was very dark even with a baked light directly above it. In order to optimize the light for this purpose, the light had to stay baked, and this was not a problem that could just be solved by going from a baked light to a real time light. So the lighting setup around the console was changed and tweaked for the console to react more appropriately to the baked lighting in the scene. To improve the console lighting while keeping the overhead light baked, Light Probes were configured around the console area. The lighting was then re-baked so that the console could receive more appropriate baked lighting information. Once the lighting setup was set up, the scene light was re-baked to create new lighting information. This resulted in the console being better lit and matching the rest of factory lighting so that it was easier to see and looked consistent with the other factory lighting. This makes the console and controls more visible and also keeps the performance benefits of baked lighting in the VR simulation.

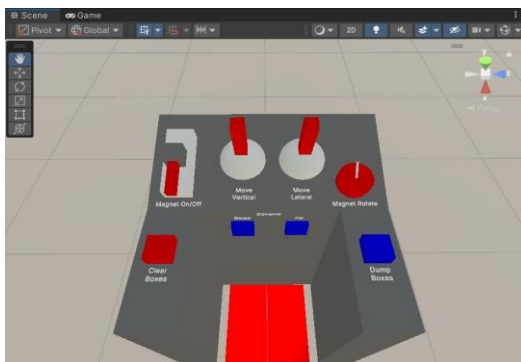


Figure 2.5.1:

Console before lighting repair

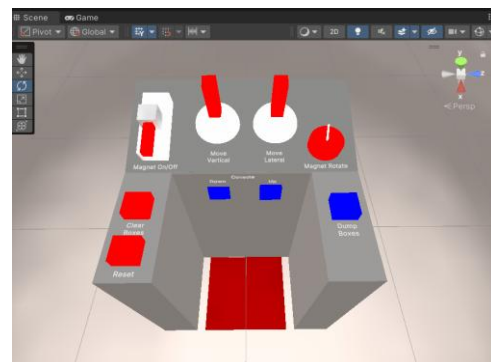


Figure 2.5.2:

Console after lighting repair

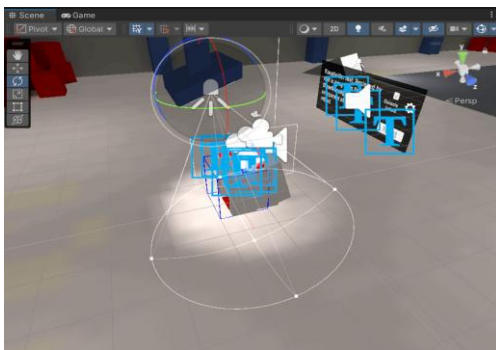


Figure 2.5.3:

Console light spot configuration

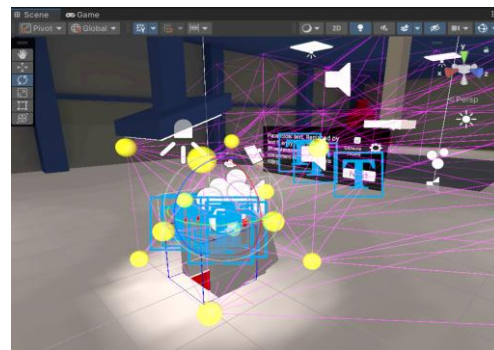
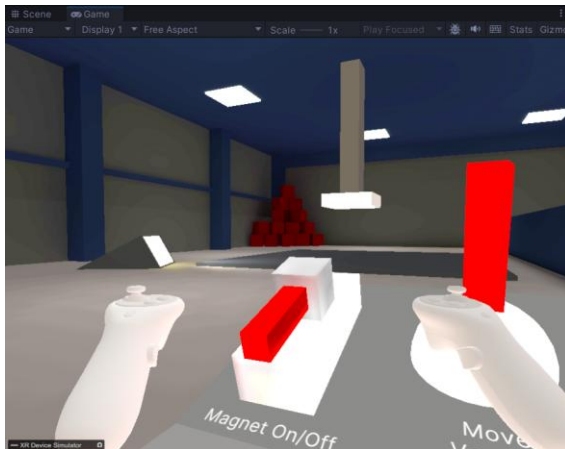


Figure 2.5.4:

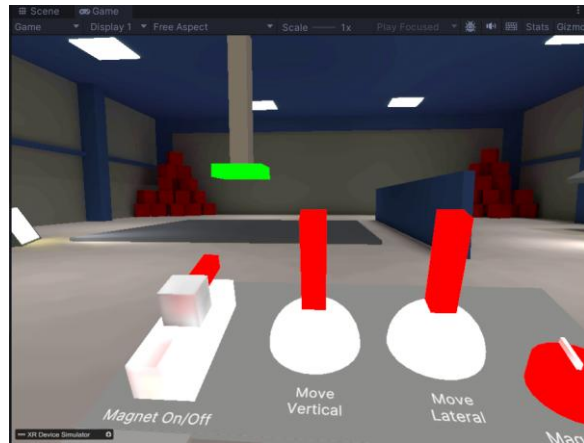
Add console light probe

2.6 Visual Feedback — Magnet State

The sixth repair aimed to obtain an unambiguous visual indication of factory magnet operation. The original prototype did not have a visual indicator to indicate whether the magnet was on or off, which may result in confusion when using controls to operate the magnet. The visual state of the magnet was set to be altered depending on its status so as to improve interaction. Once the magnet is switched on, the magnet turns from white to green to give instant visual feedback of the magnet is switched on and ready to interact with the boxes. Its look changes when it is not functioning, to make it easier for the user to tell the difference between the active and inactive states. The colour change is complemented by the necessary particle effect to indicate the activity of the magnet when lifting a box. The particle effect also helps to make the active interaction more apparent, giving the user a secondary cue that the magnet is actively interacting with an object. The result is the combination of colour-based state feedback and particle effects allows the magnet to be more easily understood and operated, reduces uncertainty when interacting with the magnet, enhances the overall responsiveness and visual quality of VR factory training.



*Figure 2.6.1:
Magnet in inactive state*



*Figure 2.6.2:
Magnet ON with green visual indicator*

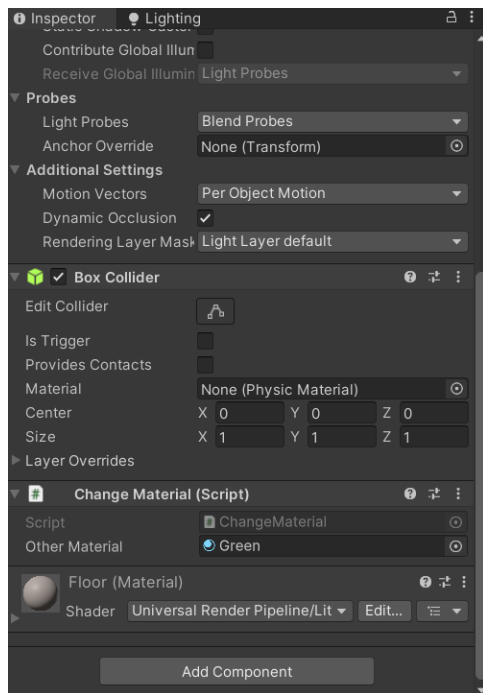


Figure 2.6.3:

Material Change Script Configuration

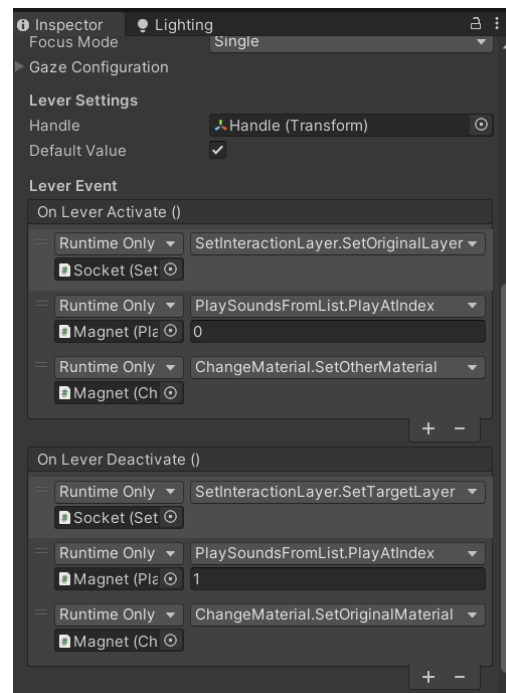


Figure 2.6.4:

Material Change Logic Configured for the Magnet Lever

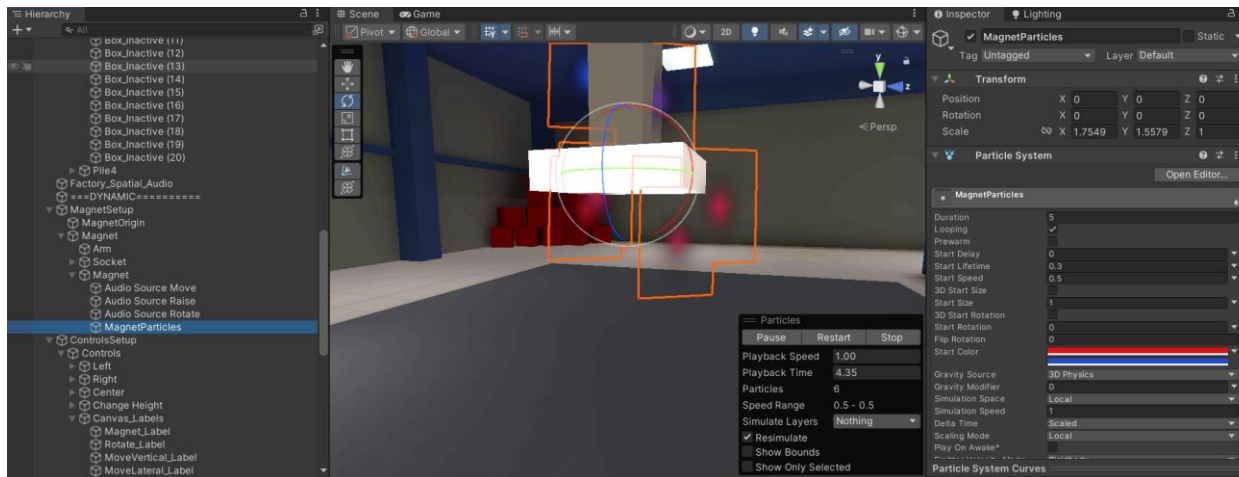
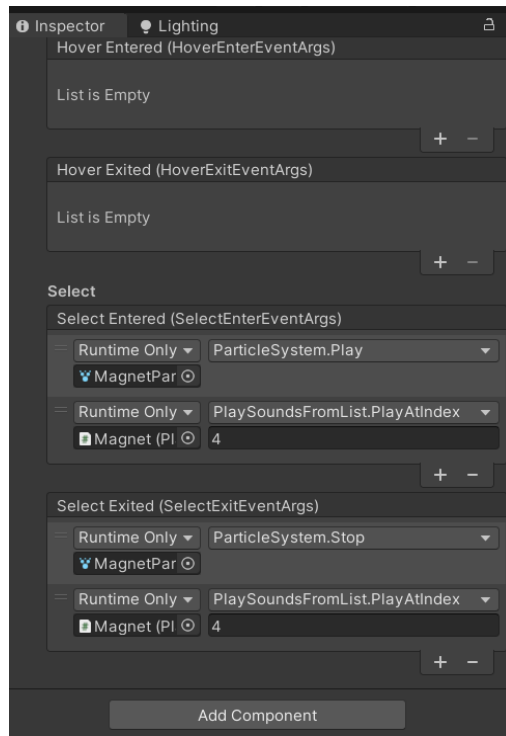
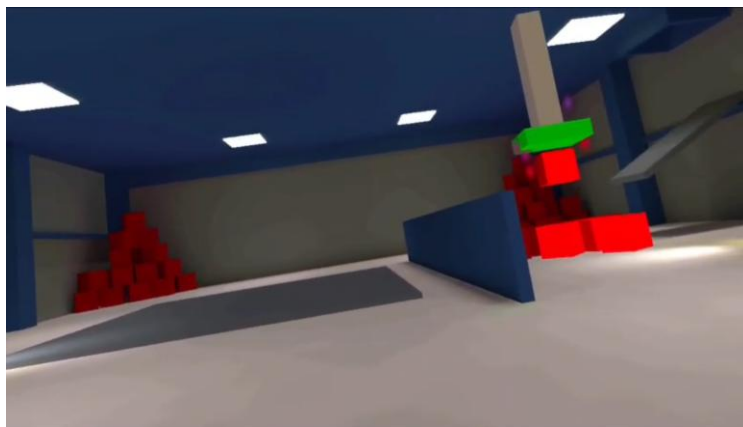


Figure 2.6.5:

Magnet particle system configuration



*Figure 2.6.6:
Configure magnet particles logic into magnet socket*



*Figure 2.6.7:
Particle effect while the magnet is actively lifting a box*

2.7 Interaction — Instruction Lock

The 7th repair was to ensure that the user can't interact with the factory controls until they finish the onboarding instructions. In the initial prototype the ray and direct interactors remained in place and the onboarding instructions were shown, so that the user can interact with the console before completing the necessary sequence of instructions. This may lead to unintended movements and disrupt the flow of training. To overcome this, the interaction settings related to the onboarding process were changed to deny access to the factory controls during the instructions. The controls were set up with the relevant XR interaction settings and the onboarding interaction logic was tweaked to enable/disable the corresponding XR interaction. When the user is new to the machine, he can't work the controls of the factory so the instructions can be given without unwanted interaction. Once the onboarding sequence is finished, the interaction state is set so that the user can continue with the factory training activities in accordance with the design. This repair makes the transition from the instruction stage to the interactive training stage clearer and lets users avoid operating the equipment at the factory until they get the instructions they need.

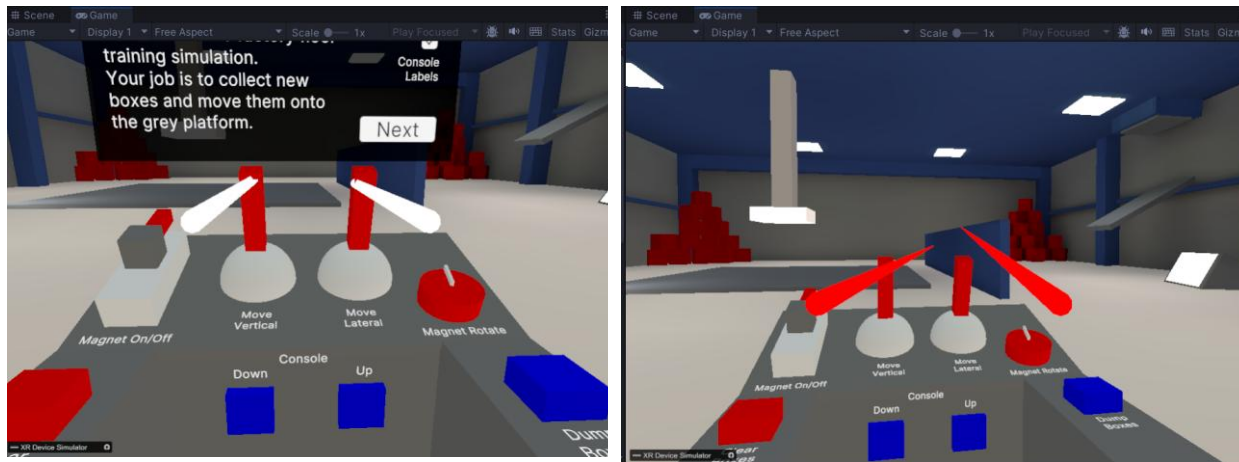


Figure 2.7.1:

*Able to access ray and direct interactors before and after reading the instruction
(Before repair)*



Figure 2.7.2:

Able to access ray before reading the instruction (After repair)

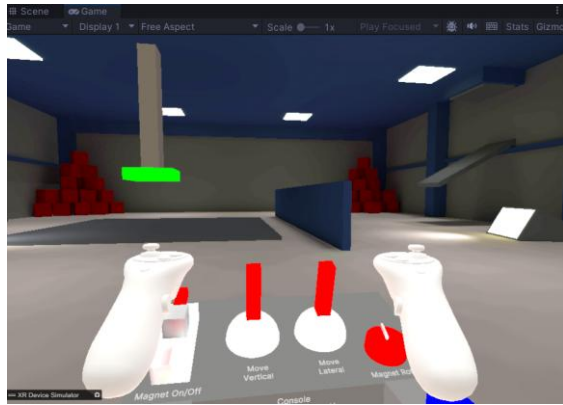


Figure 2.7.3:

Unable to access ray after reading the instruction (After repair)

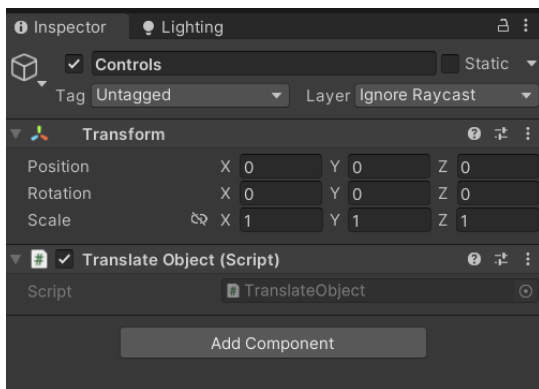


Figure 2.7.4:

Set controls layer to ignore raycast

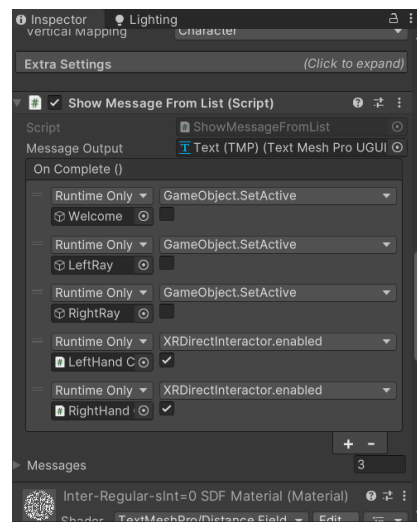


Figure 2.7.5:

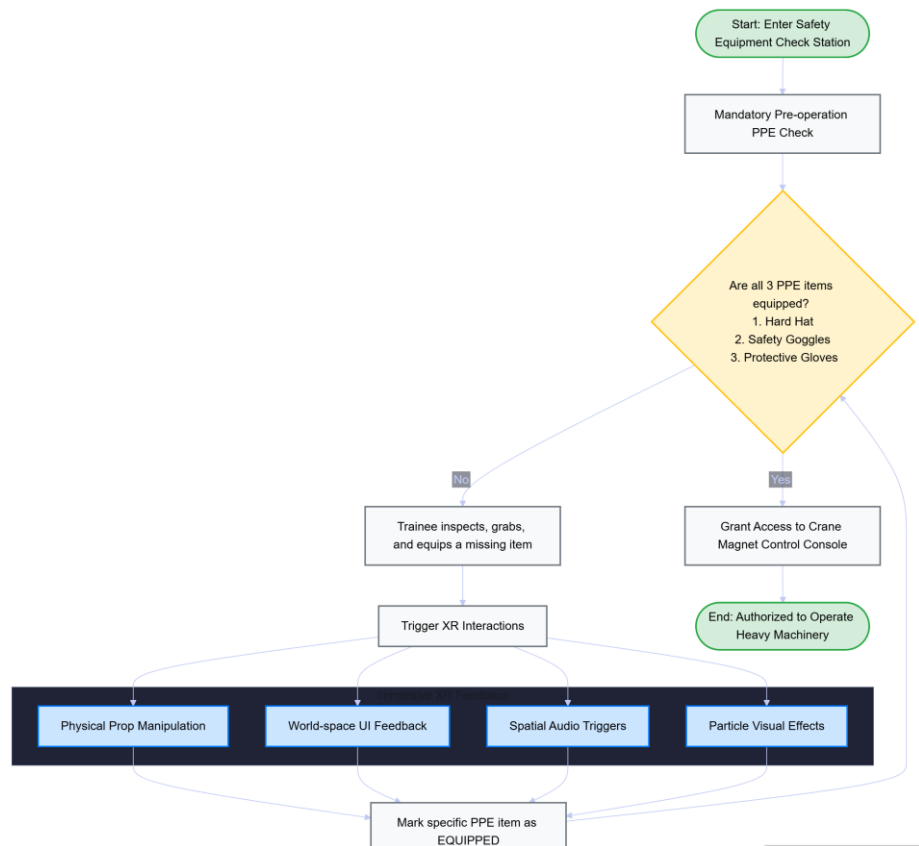
Interaction lock configuration in canvas main text

3.0 Part B

3.1 Feature Overview

In industrial workplace environments, personal protective equipment (PPE) compliance is an essential safety requirement before personnel are authorized to operate heavy machinery. To expand the factory training prototype with meaningful and immersive XR interactions, the Safety Equipment Check Station was implemented as our custom Part B feature.

This module functions as a mandatory pre-operation checkpoint where the trainee must inspect, grab, and equip three essential safety items, which are a Hard Hat, Safety Goggles, and Protective Gloves, before access to the crane magnet control console is granted. The design reinforces workplace safety compliance while demonstrating core XR principles including physical prop manipulation, world-space UI feedback, spatial audio triggers, and particle visual effects in response to the player's actions.



3.1.1 Personal Protective Equipment flowchart

3.2 Station Setup

The Safety Equipment Check Station is placed near the console's zone on the factory floor to establish a logical training sequence before being able to access the Canvas console.

1. Station Table

A dedicated workbench model was placed with designated zones for each personal protective equipment.

2. Interactive Props

High-visibility custom 3D models representing the hard hat, goggles, and gloves were created and set up with appropriate colliders as well as XR Grab Interactable components, allowing the player to pick them up naturally using direct or ray interactors.

3. Physics and Bounds

Each prop was equipped with a Rigidbody and collision boundaries in order to ensure weight and response when handled or placed on the table.

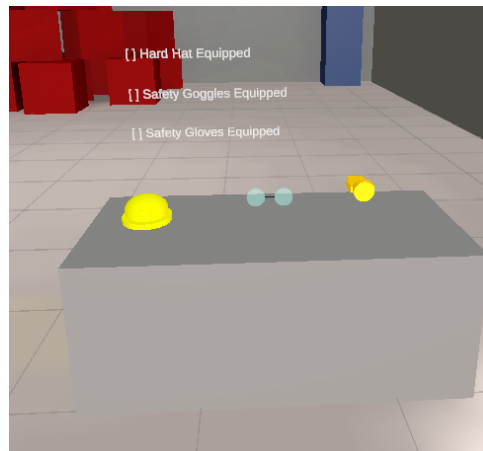


Figure 3.2.1 Station Table with the PPE Items and Checklist

3.3 World Space UI

To communicate training progress clearly without cluttering the user's field of view with fixed HUD elements, a World Space Canvas checklist is positioned directly above the station table.

1. Dynamic Task Feedback

The panel displays real-time status entries for each PPE item:

- a. ☐ Hard Hat Equipped switches to ☒ Hard Hat Equipped with the bracket and OK turning green upon player equipping
- b. ☐ Safety Goggles Equipped switches to ☒ Safety Goggles Equipped with the bracket and OK turning green upon player equipping
- c. ☐ Safety Gloves Equipped switches to ☒ Safety Gloves Equipped with the bracket and OK turning green upon player equipping



Figure 3.3.1 Task Feedback Upon Equipping All PPE Items

2. Authorization Status

To enforce safety compliance on players before machinery operation, an access barrier is linked between the controls console along with the safety station and the Canvas console. This serves as a visual blockade so that the player is unable to fully enjoy the experience, making sure the player will equip these PPE items first before being able to move and rotate the magnet and boxes.

3. VR Readability

Font sizing, contrast, and distance settings were calibrated to ensure crisp legibility inside the headset from standard standing distances.

3.4 VFX and Audio

Multi-sensory cues were implemented to give immediate feedback for correct and incorrect trainee actions, such as:

1. 3D Spatial Audio

An Audio Source component configured with full 3D Spatial Blend (1.0) and realistic linear falloff curves was attached to each Socket responsible for dictation of where each PPE Items will attach to. When an item snaps into place, a “Ding” sound is played at the socket’s position to enable a more immersive experience. The exact Audio used was “SFX_UI_Game_Positive”.

2. Visual Particle Effects

A dedicated particle system is triggered upon each item’s equipping action, emitting a celebratory burst effect over where the sockets are. This also helps communicate with the player where these items are connected to visually. This is handled in Success Particles.

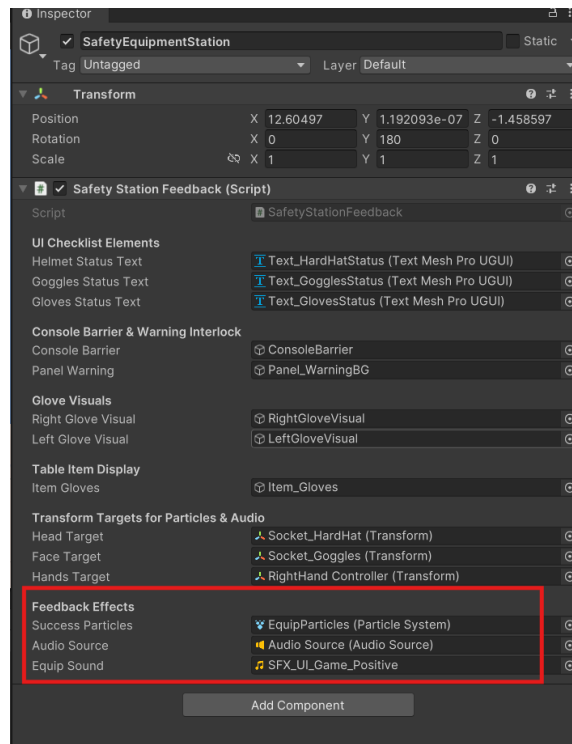


Figure 3.4.1 Success Particles and Audio Source settings

3.5 Prefab Assembly

To keep project hierarchy clean, the GameObject SafetyStationFeedback was pushed to be the parent of the PPE items, EquipParticles, Audio Source, and ConsoleBarrier. There also exists a counterpart of it packed into a Prefab, which is hosted inside an Asset Folder

PartB_SafetyStation. The Folder also hosts the materials applied to each of the PPE items, the scripts AutoSnapToSocket and SafetyStationFeedback, as well as an XR Rig Prefab for any errors relating to buggy or broken Game views. The scripts mentioned were completed with the help of AI-Assistance.

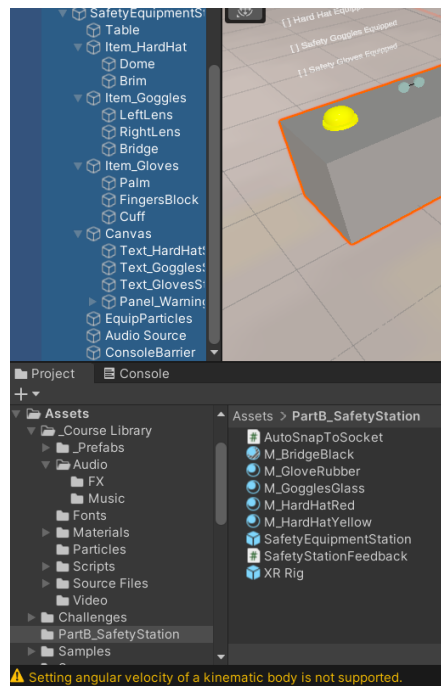


Figure 3.5.1 Assembled Hierarchies and Prefabs

3.6 Safety Equipment Sockets & Interaction Logic

3.6.1. XR Grab and Socket Mechanics

The core of the feature relies on the Unity XR Interaction Toolkit. Each piece of PPE (Helmet, Goggles, Gloves) was configured with an XRGrabInteractable, a Rigidbody for proper physics, and specific Colliders.

To ensure users could not equip items incorrectly (e.g., placing gloves on the head), we utilized Interaction Layer Masks. Each socket (XRSocketInteractor) was assigned a unique layer (e.g., "HelmetLayer") that only permitted interaction with the corresponding PPE object, preventing logical errors during the simulation.

3.6.2. Contextual UI and Audio/Visual Feedback

A central script, `SafetyEquipmentStation.cs`, manages the state of the equipment. We utilize event listeners (`selectEntered` and `selectExited`) on the `XRSocketInteractors` to trigger logic whenever an item is equipped or removed.

- **Visual Feedback:** Upon the successful socketing of all three items, a Particle System instantiated near the doorway emits a celebratory spark/glow effect, fulfilling the Part B minimum requirement for particle effects at key interaction moments.
- **Audio Feedback:** An `AudioSource` triggers a satisfying click/success sound upon each successful socket attachment, reinforcing positive user action.

3.7 Station Reset Functionality

Fulfilling the requirement for a seamless reset system, we implemented a dedicated script method bound to a physical button within the VR environment. When triggered, the system executes the following logic to restore the initial state without invoking `SceneManager.LoadScene`:

1. **Socket Ejection:** The script verifies if any `XRSocketInteractor` currently holds an object. If true, it programmatically forces the socket to release the `XRGrabInteractable`.
2. **Transform Restoration:** The script caches the initial `Vector3` position and `Quaternion` rotation of the Helmet, Goggles, and Gloves during the `Start()` method. Upon reset, the items' transforms are snapped back to these cached coordinates on the station desk.
3. **Physics Zeroing:** To prevent items from retaining momentum and flying off the desk upon reset, the script accesses each object's `Rigidbody` and sets both the velocity and `angularVelocity` to `Vector3.zero`.
4. **UI and Barrier Reset:** Finally, the script re-evaluates the equipment state (which is now empty), reactivates the Warning UI, resets the checklist to red text, and reactivates the factory door barrier.

Appendix

Team Contribution Summary

Part A - Repair, Lighting & Optimization	Yeo Pei Han	Tee Ing Harn	Leong Sum Yue	Teoh Wei Zhong
1) UI — Onboarding Fade Screen			✓	
2) Toggle Logic — Console Labels			✓	
3) Lighting — Factory Floor			✓	
4) Lighting — Box Piles			✓	
5) Lighting — Console				✓
6) Visual Feedback — Magnet State				✓
7) Interaction — Instruction Lock				✓
Additional Features				
8) Locomotion				✓
9) Comfort Settings				✓
10) Spatial Audio			✓	
11) “Reset” Button			✓	✓
Part B - Additional Interactive Feature (Safety Equipment Check Station)				
1) Station Setup — 3D Models & Props		✓		
2) World Space UI — Checklist Panel		✓		

3) VFX & Audio — Success Effects and Spatial Audio		✓		
4) Prefab Assembly — Package Station Components		✓		
5) Safety Equipment Check — Helmet, Goggles & Gloves	✓	✓		
6) Station Reset — Restore Initial State	✓			

AI-Assisted Development Declaration

This project was developed with the support of Artificial Intelligence (AI) tools, such as ChatGPT. The primary use of the AI assistance was to troubleshoot problems with Unity, offer suggestions to the programmer, and enhance a specific block of C# code. AI support was also utilised to develop and troubleshoot the Reset function, including the Training Reset.cs script to reset object positions, rotations, reset Rigidbody movement and interact with objects linked to the Rigidbody. Other than that, other AI-assisted scripts include AutoSnapToSockets.cs and SafetyStationFeedback.cs. The code and solutions were verified, tested and adjusted by the team to make them work in the VR factory environment. The use of AI tools was also assisted with the wording and organization of project documentation. All text and code generated through the use of AI has been reviewed and verified by the team, and the team will carry the ultimate responsibility for putting it into practice and submitting it.